Corpus studies and functional linguistic theories*

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This paper discusses the relationships between corpus linguistics and functionalist theories, specifically in the light of the distinction which has been proposed between 'corpus-based' and 'corpus-driven' approaches. It argues that functional theories must take on board the findings of corpus-driven linguistics if they are to fulfil the aims they set for themselves. It also presents a selective survey of corpus work in two functional theories, Functional Grammar and Systemic Functional Grammar, and concludes that although much valuable work has been and is being done to make particular aspects of lexicogramamatical description more sensitive to attested linguistic data, more should be done to test the fundamental theoretical claims of such theories rigorously against what corpora can tell us, these claims being modified or even abandoned where necessary.

1. Introduction

In this paper I shall be concerned with the relationships between corpus linguistics and functionalist theories. Formalists have generally eschewed the use of corpus data, but there are, I shall argue, good reasons why functionalists should make full use of it. Interestingly, although many functionalists routinely use textual materials in their studies, it is only quite recently that they have begun to take full advantage of available computer-readable textual data. Furthermore, the use of corpora which has so far been made by functional linguists has come in for some quite serious criticism by those who believe that we should build up descriptions, and ultimately a theory, based on what corpora tell us, approaching the corpus data with as few theoretical preconceptions as possible. The aim of this paper is to explore these positions and suggest ways forward.

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2. Formalism, functionalism and linguistic data

Chomsky (1965: 4), as is well known, claims that the study of the observed use of language cannot constitute the subject matter of linguistics if this is to be seen as a serious discipline: he is interested in linguistic **competence** rather than **performance**, and argues (1965: 18) that information about competence cannot be extracted from data by inductive procedures. Although he goes on to observe that performance data can provide evidence for the evaulation of hypotheses about competence, in practice formal linguistics has been based primarily on the intuitions of the native speaker (often the linguist him/herself), a procedure which Chomsky (1965: 20) himself clearly favours.

In later work (especially *Knowledge of Language*, Chomsky (1986)), the competence/performance distinction is recast in terms of that between **Externalised Language** (**E-Language**), as available in corpora of language use, and **Internalised Language** (**I-Language**), the form in which language exists in the brains of human beings. This underlines even more strongly the relevance, for formal linguistic theory, of native speaker introspection and the rejection of performance data as primary evidence. Although, in a yet more recent discussion, Chomsky (2000) stresses that no types of evidence for underlying linguistic competence can be ruled out in advance, he nevertheless remains sceptical about the feasibility of a theory of language use.

Chomsky's formalist position contrasts dramatically with that of Sinclair, who has perhaps been more persuasive than any other linguist in his advocacy of an approach to language based on corpora:

> …we should strive to be open to the patterns observable in language in quantity as we now have it. The growing evidence that we have suggests that there is to be found a wealth of meaningful patterns that, with current perspectives, we are not led to expect. We must gratefully adjust to this new situation and rebuild a picture of language and meaning which is not only consistent with the evidence but exploits it to the full. This will take some time, and the first stage should be an attempt to inspect the data with as little attention as possible to theory. (Sinclair 1992: 6)

Sinclair's approach to language is an essentially functional one, as confirmed by his observation that

> The new evidence is just beginning to raise queries about the relation of grammar to meaning, and is supporting the growing movement towards functional grammars and lexicogrammars. (Sinclair 1991a: 489)

Let us explore in a little more detail the essential differences between formalism and functionalism which make corpus linguistics of so much more relevance to the latter than to the former.

Formal theories of linguistics, exemplified here by those of Chomsky, are typically concerned with what the study of language can tell us about the structure of the human mind, and so tend to play down the communicative functions of language:

> Human language is a system for free expression of thought, essentially independent of stimulus control, need-satisfaction or instrumental purpose. (Chomsky 1980: 239)

Functional theories, on the other hand, take it as a centrally important fact that the primary function of language is communicative:

> The essence of functionalism [...] is that language meets communicative ends. (Nichols 1984: 101–102)
>
> The primary aim of natural languages is the establishment of inter-human communication; other aims are either secondary or derived. (Dik 1986: 21)
>
> In the functional paradigm, [...] a language is in the first place conceptualized as an instrument for social interaction among human beings, used with the intention of establishing communicative relationships. Within this paradigm one attempts to reveal the instrumentality of language with respect to what people do and achieve with it in social interaction. (Dik 1997a: 3)
>
> The theme underlying the various functional approaches is the belief that language must be studied in relation to its role in human communication. Language is thus viewed as a system of human communication, rather than as an infinite set of structural descriptions of sentences. (Foley and Van Valin 1984: 7)

We may conclude that while formal linguistics has little place for the study of actual linguistic usage, functional linguistics should, if it is true to its precepts, take very seriously the new evidence which corpora make available.

In practice, the distinction between formalism and functionalism is not a straightforward dichotomy; rather, there is a cline from Chomskyan models at one end, to radical functionalist proposals such as the 'emergent grammar' of Hopper (1987, 1992) at the other. In Butler (2003) I propose that between the two extremes we may recognise theories such as Role and Reference Grammar (RRG: Van Valin and LaPolla 1997, Van Valin in press) and Functional Grammar (FG: Dik 1997a, 1997b) towards the more formal end of the cline, and Systemic Functional Grammar (SFG: Halliday 1985, 1994a, 1994b; Halliday and Matthiessen 1999, 2004) and various strands of what is sometimes known

as West Coast Functionalism[1] (WCF: e.g. Givón 1989, 1993a, 1993b, 1995, 2001a, 2001b) towards the most radically functional end. We might expect that proponents of theories near the most radically functional end of the spectrum would be particularly committed to the study of authentic language productions.

Before we proceed any further, a final comment needs to be made about the respective roles of introspection and attested data in relation to theory building. Even those, such as Sinclair, who advocate an uncompromisingly usage-based approach recognise that the analyst's own introspective judgments are bound to play a part in linguistic description and theorising, if only because the evidence from corpora still needs to be interpreted, and this activity inevitably brings in what the linguist already knows and believes about language. The important distinction here is between the use of introspection **as data** and the essential utilisation of our intuitions to formulate hypotheses and analyse data (see Rundell 2001, cited in a discussion by Stubbs, forthcoming). It has been shown time and again that introspection is a very unreliable way of getting at details of how people really behave linguistically, so that if our aim is to account for real usage, then the use of introspection to furnish primary data is clearly seriously flawed. For further discussion of these issues, see Chafe (1992), Fillmore (1992), Aarts (1991, 2002), Sinclair (2004).

3. What is a corpus?

The term 'corpus' could logically be applied to any body of texts, however small and homogeneous, which the linguist assembles for the purpose of analysis, or obtains from some other source. Indeed, this is the sense in which the term is sometimes used by those who, although they might want to claim that they do 'empirical' or 'data-intensive' linguistics rather than 'corpus linguistics', do base at least some of their work on textual studies. This is not, however, the sense of the term which is intended nowadays when we speak of corpus linguistics. The definition of a corpus given by EAGLES (the Expert Advisory Group on Language Engineering Standards) is as follows:

> A *corpus* is a collection of pieces of language that are selected and ordered according to explicit linguistic criteria in order to be used as a sample of the language. (EAGLES 1996)

Thus a corpus, in this sense, must consist of pieces of authentic language, and these pieces of language must be selected according to well-defined criteria, in

order to make this sample of the language (or, we could add, of a variety of a language) representative of the population, so increasing the confidence with which we may extrapolate findings from the corpus to the language or variety as a whole. The criteria for selection and classification of corpus materials have so far been largely external — the nature of the participants and of the occasion of use, the social setting, the communicative function which the pieces of language serve, and so on. Work has also been done on criteria for ensuring that sample sizes are large enough to provide stable distributions of a range of internal, linguistic features. However, as Sinclair (2003, forthcoming) rightly points out, it would be inadvisable to use internal criteria as the basis for text selection for a corpus, since this would lead to circularity: if the compilers of a corpus include a particular set of texts because, in their view, they illustrate the typical linguistic features of such texts, then we cannot know whether the 'discovery' of a particular feature in the texts is a genuine reflection of the properties of that variety, or merely one of the features which led the compilers to choose those particular texts.

A corpus thus differs fundamentally from a **collection** or an **archive** of textual material in that the latter are not necessarily ordered, the selection of material need not be made on linguistic criteria, and there may be no claim that the material is in any sense representative of a language, or one of its varieties.

There are also some other features which, though they do not form part of the set of minimal criteria given above, also serve to distinguish corpora from other bodies of text. Firstly, modern corpora are **large** — and what is meant by 'large' has changed considerably over the 40 years or so since the compilation of the first corpora to conform to the definition given earlier. When Kučera and Francis began their work on the Brown Corpus of written American English in 1961, their target of 1 million words seemed huge; nowadays, corpora such as the British National Corpus (henceforth BNC) and the COBUILD Bank of English (henceforth BofE) are in the nine figure area. Secondly, the large size of corpora makes it necessary to search them using automated tools, so that a further feature of the modern corpus is that it is **computer-readable** and is interrogated using software tools which may be commercially or even freely available, or developed in-house for particular projects. Thirdly, whereas text collections and archives tend to consist of complete texts, many corpora contain **text fragments** of particular sizes — e.g. the Brown Corpus and the Lancaster-Oslo-Bergen (LOB) Corpus which is parallel to the Brown but samples written British English from the same year, 1961, both consist of 2000-word samples drawn randomly from a set of texts chosen to represent 15 text types. EAGLES,

however, has recommended the use of whole texts, presumably in order to facilitate work on areas of discourse/text structure for which the entire text is required. Such a recommendation is clearly in line with a functional approach in which a text is seen as the record of a communicative event.

It is important to recognise, however, that just as with our consideration of functional and formal linguistics, we shall find that in practice the distinction between corpora and other kinds of text collection is not as clear as we might hope for: much depends on the degree of systematicity of the selection process.

Representativeness is what makes corpora, as defined above, particularly useful: it is, however, one of the more difficult aspects of corpus design. The corpus compiler must decide what weight to give to the various factors involved: the balance of spoken vs. written language, the communicative functions which are to be represented, the importance to be attached to the perceived popularity or social influence of particular types of text, the ease or difficulty of obtaining (and, if necessary, transcribing) the materials, and so on (see Atkins, Clear and Ostler 1992; Biber 1993a, 1993b). Prospective compilers could do no better than follow the practical guidelines set out by Sinclair (forthcoming): decide on the structural criteria to be used, and use them to create a framework for building the main corpus components; list the text types found in each component, using only external criteria; prioritise these text types according to a set of motivated weighting factors; estimate the size required for each text type in each component; compare the corpus with the original plan at all stages; and document all the steps taken in corpus design. This last recommendation is crucial in enabling users to assess the appropriateness of a corpus for their purposes, and in helping them to account for unexpected findings.

If the corpus really does represent the language or variety under study, we may be confident, statistical factors permitting, that our results have a validity which transcends the corpus itself. Size is also an important consideration: clearly, the larger the corpus, the more data it is likely to provide about particular linguistic features, and the more confident we can be of the validity of our findings. Indeed, if the events under focus are rare, even a very large corpus may yield only a small set of results; on the other hand, where the phenomena under study are frequent, a smaller corpus may suffice. The conclusion we may draw, then, is that if our aim is to give an accurate description of some particular area of a given language or language variety, based on authentic data, and/or to test hypotheses as part of the process of theory building, then it is to a representative corpus that we should turn if possible, rather than to any other type of text collection. If our interest is in comparison across languages, the same principle

applies: we need corpora of two or more languages which are as similar as possible in their design, either because one corpus contains translations of the texts in the other (i.e. **translation** corpora, sometimes referred to as **parallel** corpora), or because the external criteria chosen for selection of the material are the same or at least similar (**comparable** corpora). Furthermore we should use a corpus which is large enough to provide an adequate set of examples, and this, for many types of study, entails the use of a large corpus and so brings with it the need for computer-assisted analysis.

With these distinctions in mind, we now turn to the role of the corpus in linguistic description and theory building, as a necessary preliminary to assessing the current state of corpus-oriented work in functional linguistics.

4. The challenge to functional linguistic theories from corpus-driven linguistics

Corpus linguistics, like functional linguistics, comes in various flavours. Tognini-Bonelli (2001) recognises two broad types, which she calls **corpus-based** and **corpus-driven**. The term 'corpus-based' is used

> …to refer to a methodology that avails itself of the corpus mainly to expound, test or exemplify theories and descriptions that were formulated before large corpora became available to inform language study. (Tognini-Bonelli 2001: 65)

She sees this approach as reflecting linguists' confidence in the models they espouse, and the data on which they are based: the corpus is used to refine the model, providing a quantitative dimension, rather than to challenge existing categories and ways of thinking about language. Tognini-Bonelli (2001: 68–77) suggests three ways in which these linguists have reacted to data which prove uncomfortable for their theories.

Firstly, linguists may attempt to insulate their theories by relegating the data to secondary status with respect to theoretical statements — e.g. decide that the corpus data pertain only to performance, and do not affect competence.[2] Secondly, the raw data may be systematised by attaching word tags or even syntactic parse trees. Tognini-Bonelli (2001: 72–74) points out that although tagging can indeed be useful, the tag categories are initially provided by the linguist rather than derived purely from the corpus, and so even when taggers are trained on further corpus material, the ultimate product is an annotation system which imposes a particular view of the data. Indeed, Sinclair (1997: 22)

writes of "insisting that word-classes have to be completely rethought in the light of corpus evidence of the similarity of words in their corpus patterns", though he also insists that this does not mean wholesale rejection of the 'existence' of categories such as noun and verb. Thirdly, under the heading of 'instantiation', Tognini-Bonelli discusses the use of corpus analysis to provide probabilities for the occurrence of particular types of linguistic item and structure, chosen by the writer or speaker from the range provided by the language in question.

The corpus-driven approach, on the other hand, is characterised as follows:

> In a corpus-driven approach the commitment of the linguist is to the integrity of the data as a whole, and the descriptions aim to be comprehensive with respect to corpus evidence. The corpus, therefore, is seen as more than a repository of examples to back pre-existing theories or a probabilistic extension to an already well defined system. The theoretical statements are fully consistent with, and reflect directly, the evidence provided by the corpus. (Tognini-Bonelli 2001: 84)

In other words, we are invited, as Sinclair (1992) puts it, to 'trust the text', deriving our descriptions, and ultimately our theoretical statements, directly from the corpus data, with a minimum of interference from preconceived ideas about the lexicogrammar, as reflected, for example, in the annotations included in tagged or parsed corpora.

There can be no doubt that the corpus-driven strategy has proved extremely fruitful in unearthing many fascinating insights, mainly though not exclusively concerned with English. What follows is a necessarily brief summary of the main lessons we can learn so far from this approach.

The most important finding is that lexical and grammatical patterning, and also meaning and form, are intimately bound up together — so much so, in fact, that in the view of these linguists there is no sense in separating them. This conclusion has been reached through detailed studies of the patterning of particular node items, as revealed by concordances produced from raw text. Crucial here are the syntagmatic relations of **collocation** (co-occurrence between 'lexical words') and **colligation** (originally coined by Firth to mean the co-occurrence relationship between grammatical categories such as noun and adjective, but now used more widely to cover relationships between grammatical categories and particular lexical words as well). Studies of such patterns have given considerable weight to the postulation of a unit, the **lexical item** or **extended unit of meaning** (see especially Sinclair 1996, 1998; Tognini-Bonelli 2001).

We may illustrate the concepts involved by reference to Sinclair's (1996) analysis of an item centred around the words *naked eye*. The **core** of this item consists of the irreducible minimum found in all occurrences of the composite unit, in this case simply the collocation of *naked* and *eye*, whose meaning is not deducible in any simple way from that of the individual words. At the first position to the left of this collocation, the word *the* occurs in 95% of the 151 distinct concordance lines from the BofE, which at the time contained 211 million words. The remaining 5% can be explained in terms of other regular grammatical processes in English. At the second position to the left, we have a pattern of colligation: prepositions occur here in 90% of cases, *with* and *to* being the most common. The third position to the left is dominated by the words *see/seen* and *visible/invisible*: we can see here both collocation, in that particular lexical words are involved, and colligation, in that only verbs and adjectives are involved. More importantly, however, we have here a clear **semantic preference** for words concerned with visibility; indeed, this preference spills over into the fourth position to the left of the core too. Furthermore, there is a correlation between the colligation with verb or adjective, and the appearance of *with* or *to*, respectively, as the preposition at the second position to the left. Finally, examination of the wider context of the item reveals what is known as a **semantic prosody** (sometimes called 'discourse prosody'), a meaning of an attitudinal kind, in this case one of difficulty, present in over 85% of the examples, and realised not only through the lexical negative *invisible*, but also through words such as *small*, *faint*, *weak*, *difficult* with *see(n)* and *barely*, *rarely*, *just* with *visible*, or hinted at by even subtler means such as the modals *can* or *could*. The lexical item proposed in this case, then, is 'visibility + preposition + *the* + *naked* + *eye*', plus the overall semantic prosody of difficulty. Many more examples of semantic preferences and prosodies can be found in the publications by Sinclair and Tognini-Bonelli cited above, and also in the work of Stubbs (1995, 1996, 2001), who develops these ideas in terms of the light they can shed on the isolation of significant **cultural keywords**.

So pervasive are the kinds of pattern I have just illustrated that Sinclair proposes they constitute the norm in language use. He distinguishes between two types of principle governing the use of language. One is the **open-choice** principle, according to which sentences are created on a kind of slot and filler basis: the grammar generates the slots and then any item which fits syntactically and semantically into a given slot may, in principle, be used. This, of course, is the principle on which grammars, including functional grammars, have so far been formulated. The evidence from corpus-driven linguistics, however,

suggests that this principle accounts very poorly for the patterns found in actual language use, and that a second type, the **idiom principle**, is dominant:

> The principle of idiom is that a language user has available to him or her a large number of semi-preconstructed phrases that constitute single choices, even though they might appear to be analysable into segments. (Sinclair 1991b: 110)

Note particularly the last part of this statement: many of the phrases which are held to be semi-preconstructed, and so to act as unitary choices, could perfectly well be generated by the open choice grammar, but in normal language use are not.

Further convincing evidence for the importance of (semi-)preconstructed 'formulaic' sequences has come from work within a very different perspective: Wray (1999, 2000, 2002; Wray and Perkins 2000) has developed a sophisticated socio-psycholinguistic model of formulaic language on the basis of data from non-disadvantaged and aphasic adult language and the language of first and second language learners. Wray proposes

> …that it is the accessing of large prefabricated chunks, and not the formulation and analysis of novel strings, that predominates in normal language processing (Wray 2002: 101)

and that analysis and generation according to the open choice grammar occurs just on a needs only basis, under conditions where holistic processing is insufficient.

There are other, related features which emerge from the corpus-driven type of study. It is normal, in a dictionary, to find words listed as lemmas, the convention being that what is listed is the infinitive of a verb, the singular of a noun, the basic (non-compared) form of an adjective, and so on. This reflects, and in turn fosters, the assumption that it is the lemma, a distillation from the actual forms of a word, which carries lexical meaning and patterns with respect to other words. Corpus-driven studies, however, have shown this to be an unjustified assumption in many cases, in that the different forms of, say, a verb may have different preferred senses and different collocational and/or colligational patterns (see Sinclair 1985: 91 on forms of DECLINE, Sinclair 1992: 13 on MOVE, Stubbs 1996: 172–173 on EDUCATE, Butler 1998: 185 on CONSIDER, Tognini-Bonelli 2001: 96–98 on the alternative infinitive forms *saper* and *sapere* in Italian). A related point is that a word which may be used as two different parts of speech may show different patterning in the two types of environment (see Sinclair 1992: 14 on the nominal and verbal forms of *combat*, which tend to refer to physical and social fighting, respectively). Furthermore, some words

have a preference for particular syntactic environments (e.g. *lap* occurs predominantly in nominal complements of prepositions: Sinclair 1992: 14). Some words even defy traditional classification altogether: for instance a study of *of* (Sinclair 1991b: 81–98) revealed that it only rarely performed a clearly prepositional role, more normally occurring in structures which extend prenominal structure or provide a second noun with the role of semantic head (*a lot of the houses*, *the middle of a sheet*, etc.). Finally, there is evidence that the patterning of words may differ according to the language variety in which they are being used. A clear example is provided by the use of words with special meanings in sporting registers of English: consider the items *field*, *bat*, *maiden*, *over* in the English of cricket. Differences in patterning between spoken and written registers are revealed by corpus analysis: for instance, in a study of the verb *pasar* in Spanish one particular combination, *pasar* + *lo* + evaluative adverb (e.g. *bien*, *mal*) was found in spoken texts but not in written; and the imperative use of *consider* is found with much higher frequency in academic written English than in other registers (Butler 1998: 187).

There is, then, a wealth of information which has come out of corpus-driven linguistics and which needs to be taken account of in functionalist theories. In my view, however, there are some problematic areas within corpus-driven linguistics itself, which I shall now discuss with particular reference to the concerns of functional grammarians.

Undeniably, the rich literature which has developed in the past few years on corpus analysis à la Sinclair presents a series of fascinating and detailed snapshots of individual words or groups of words, and reveals the full intricacy of the relationships between lexical and grammatical aspects of patterning in these cases. But how do we put all this together to give an overall picture? I am not, of course, trying to perpetuate the myth that we can provide a fully detailed account of all aspects of even one language, all in one go. There was, for instance, much sense in Firth's practical approach, which advocated limiting oneself to the study of particular varieties of a language. And yet, linguistic theories cannot be founded entirely on just bits and pieces here and there: we do need to find out how the parts interact to form a necessarily complex but coherent picture of at least a sizeable portion of a language.

Corpus-driven linguistics has recently begun to address this problem to some extent. Two related strands of work may be recognised here: the **local grammars** approach and the formulation of **pattern grammars**.

The idea behind local grammars arose from the realisation that even sophisticated automatic parsers cannot cope with certain areas of the grammar

of a language, which therefore need their structures describing in terms specific to the particular area, such as names and addresses, titles and amounts of money. Hunston and Sinclair (2000) link this idea to that of sub-languages, seeing local grammars as describing small sub-languages, and sub-language descriptions as extensions of local grammars. In Barnbrook and Sinclair (2001) and Barnbrook (2002) we find formulations of a local grammar for definitions, while Hunston and Sinclair (2000) discuss a local grammar for evaluations. Instead of the categories found in general grammars, local grammars employ functional categories specific to the area under description: for instance, a definition consists of something which is be defined (the definiendum), something which defines it (the definiens) and what is labelled a hinge connecting the two. Hunston and Sinclair (2000: 79) claim that the functional categories used in such local grammars are "more transparent and trustworthy" than those in general grammars, and go on to suggest that it might therefore be worth investigating how far the notion of local grammar can be extended in the description of sentences. In other words, rather than a single general grammar, we might end up with a set of local grammars for particular areas defined by their communicative functions in the discourse.

The concept of pattern grammars is explained by Hunston and Francis as follows:

> …a pattern is a phraseology frequently associated with (a sense of) a word, particularly in terms of the prepositions, groups, and clauses that follow the word. Patterns and lexis are mutually dependent, in that each pattern occurs with a restricted set of lexical items, and each lexical item occurs with a restricted set of patterns. In addition, patterns are closely associated with meaning, firstly because in many cases different senses of words are distinguished by their typical occurrence in different patterns; and secondly because words which share a given pattern tend also to share an aspect of meaning. (Hunston and Francis 2000: 3)

This approach, then, takes on board all of the proposals from corpus-driven corpus linguistics I have outlined above — the idiom principle, collocation and colligation, semantic preference and semantic prosody — but attempts to systematise the data into a grammar of patterns. The work has its roots in work by Francis showing that particular patterns are selective for words with particular types of meaning. The methodology used involves going from a pattern to items in it, and then to other patterns in which those items occur. For instance, in the pattern 'noun + *that*-clause' it was found (Francis 1993, cited in Hunston and Francis 2000: 29–30) that the nouns could be divided into semantic classes

(illocutionary processes, linguistic activity, mental states with respect to particular issues, thought processes, feelings and attitudes, general nouns). Examination of contexts in which the various nouns appear reveals that different nouns select for different semantic values in the *that*-clause, and also show further phraseological patterning (e.g. *for the simple reason that*, *it is hard to believe that*). In this way, complex networks of associative patterning can be built up. Work using this approach led to the publication of two COBUILD pattern grammars, one for the verbs in the Collins COBUILD English Dictionary (Francis, Hunston and Manning 1996), the other for the nouns and adjectives (Francis, Hunston and Manning 1998), the aim being

> …to produce lists of items that have each pattern, each list being as complete as the sample of English consulted (the 300 million word Bank of English corpus) would allow. (Hunston and Francis 2000: 32)

Hunston and Francis's book constitutes a fascinating account of what patterns are, the problems involved in identifying them, and their relationships with meaning, structure, word class and textual patterns. From the viewpoint of the functional grammarian, it is particularly instructive to look at the types of pattern statement made, and their relationship with the structural categories to be found in most grammars. The patterns are expressed in terms of formal classes such as verb, noun, adjective and adverb groups,[3] *that*, *-ing*, *to*-infinitive and *wh* clauses, together with any individual words which may form part of the pattern. For instance, the pattern **V *to* n that** (p. 46), indicates that we are focussing on the behaviour of the verb (hence capital **V**), and that it has the preposition *to* after it, followed by a noun group, followed by a *that*-clause, an example from the corpus being *Dentists have to explain to their patients that they* … Hunston and Francis (Chapter 6) explicitly choose this form of representation in preference to one involving structural functions such as Object, Complement and the like, arguing that such categories not only prove problematic to the point of making explicit decisions impossible, but also that such a representation is futile:

> In most cases, however, the structural analysis added nothing, and all that was important to say about a verb could be said in terms of its pattern and its meaning group, irrespective of the structural interpretation. Our conclusion was that structural analysis is a pointless exercise, …(Hunston and Francis 2000: 152)

The authors, perhaps aware that many readers will wonder where theory comes into all this, state firmly (p. 250) that their methodology is indeed theory-driven,

citing in support of this claim the two assumptions upon which the work is based, namely that "the behaviour of individual words is important" and that "patterning that can only be observed when the research looks at large amounts of language is important". Both statements point to orientations which an appropriate theory should have, and the second implies the important claim that the quantity of data one examines affects the quality of what one can learn from it; in my view, however, neither can be taken to justify the claim that the work is theory-driven. When addressing the question of what kind of theory might arise from corpus-driven work (pp. 259–261), they point out that their model challenges various ideas which are central to most grammatical theories: word classes, constituency, structural functions such as Object, Complement, etc., the lexis/grammar and grammar/semantics distinctions, and the role of grammar in delimiting what can and cannot be said. In place of these ideas, they would propose the integration of lexis and syntax, the close relationship between meaning and pattern, and the idea of a grammar as locating an utterance in the context of what is typically said. Furthermore, the authors adopt a stance on the role of grammars which is very much at odds with that evidenced in functional linguistics:

> …a grammar cannot explain, it can only generalise, and unifying theories may be more-or-less useful metaphors, but nothing else. (p. 260)

Corpus-driven linguists, then, wish to jettison many of the categories and distinctions made in traditional and functional grammars, and with this rejection logically comes that of the theoretical proposals built around such categories:

> Functional grammars (Dik 1968: Halliday 1985; 1994)[4] describe the internal functioning of the components of a grammar, how they relate to each other and how they create units of meaning, but it is characteristic of the meaning so produced that it remains internal to the grammar. The use of a semantically suggestive terminology indicates the area of meaning and gives a rough guide to the operational sense of a choice, but it does not reach precision of description. Words like "past", "present", "future", or "active", "passive", or even "material process", "goal", "circumstance" have this quality; words (for English) like "subject" or "object" have lost most of their relevance. Categories such as these have a vast range of use; it is thus not surprising that their application is restricted to linguistic analysis itself. (Hunston and Sinclair 2000: 79)

The implication is that the simplified labelling of a pattern grammar, and/or the functional labelling of a local grammar, provides a superior account, although Sinclair (1997: 22) recognises that "[t]he resultant theories and descriptions,

when they arise from the ashes of the old, may well turn out to be quite similar to their predecessors".

Unfortunately, the comments of the corpus-driven linguists with respect to functional grammars present a rather narrow, and so misleading, view of what such grammars are all about. Firstly, although I am not entirely clear what Hunston and Sinclair mean when they say that meaning, in a functional grammar, remains internal to the grammar, on one reading of this statement it is simply incorrect, in that functionalists attempt to motivate elements of their theories, including those relating to meaning, in terms of external pressures. Secondly, such statements do not do justice to the aims of some functional theories, including that of Dik, whose (very early) work is referenced in the above quotation. Admittedly, this may arise in part because Sinclair and his colleagues recognise that they owe much to Firth and the neo-Firthian theory of SFG championed by Halliday: the aims of Halliday's linguistics, although sharing with other functionalist approaches some basic characteristics such as the importance of language as communication and the centrality of meaning, are nevertheless very different in some other respects from those of other functional theories such as FG and RRG (for detailed discussion, see Butler 2003). In particular, the latter theories are very much concerned with the typological adequacy of their proposals, whereas such considerations have historically had a rather low priority in SFG, whose crucial developments came about very largely through the analysis of English, although recent work has given more attention to a number of other languages too. In other words, theories such as FG and RRG are concerned that the theoretical categories they set up, and the claims they make in terms of these categories, are demonstrably valid not just for English and perhaps a few other languages, but for a very wide range of languages of different types. There is, of course, a good practical reason for the dearth of corpus-driven work on languages other than English (Tognini-Bonelli's work on Italian being a significant exception): as we shall see later, large, balanced corpora of other languages have only started appearing relatively recently.

In line with their typological orientation, FG and RRG, among other theories, have given a great deal of attention to matters such as the validity or otherwise of categories such as Subject and Object to which the corpus-driven linguists object on the grounds that they are either imprecise or tell us little more than a pattern couched in class rather than functional terms. FG has strict criteria for the recognition of Subject and Object which entail that not all languages show these categories (see Dik 1997a: Chapter 10), while RRG rejects the traditional terminology, arguing that Subject is not a cross-linguistically

defensible category, and substituting a more general concept of Privileged Syntactic Argument of a construction, which, for a given language, may or may not pattern on to the Subject as seen in traditional grammars; as with FG, some languages are claimed not to have any syntactic function assignment whatever (see Van Valin and LaPolla 1997: Chapter 6). However, a grammar which did not postulate any abstract categories of this type would be quite incapable of formulating the cross-linguistic generalisations which are essential if we are to try to understand the principles which underlie language as a human capacity, rather than only the patterns evidenced in a single language.

A further important difference between corpus-driven linguistics and most functional linguistics is concerned with the cognitive dimension. Much influential work in functional linguistics is moving inexorably in the direction of greater attention to the cognitive aspects of language, and it is clear that functionalists and cognitivists share many fundamental assumptions about the nature of language, so much so that some have written of 'cognitive-functional' approaches (see e.g. Tomasello 1998, 2003, also the discussion of the relationship between the two types of approach in Horie and Comrie 2000). There is much discussion in the American functional literature of cognitive properties and constraints (e.g. in much of Givón's work), and it has been suggested that FG and Cognitive Linguistics could complement each other to form an overall framework which would be more viable than either theory taken alone (Kalisz and Kubiński 1997). The cognitive dimension is also a salient feature of the Functional Lexematic Model, which combines aspects of FG with ideas from Coseriu's Lexematics (see especially Faber and Mairal Usón 1999). In much corpus-based work, however, the cognitive dimension is all but absent. In the case of work in the Sinclair tradition, the lack of any detailed consideration of cognitive matters is understandable, in that the background of the work, in the ideas of Firth and Halliday, prioritises sociocultural interpretations of language, and itself pays scant attention to psycholinguistic matters.[5] Note, for instance, the description of Chomsky's work as "a wave of cognitive, non-textual linguistics" (Sinclair 1992: 7).

So does linguistics which is 'textual' and corpus-based (in the most general sense of the term) have to be non-cognitive? This would be a pity: if, as functional linguists, we really are committed to an account of language which provides functional motivations for why languages are as they are, then we cannot escape the need to investigate motivations which are cognitive in kind. I myself see no fundamental contradiction between sociocultural and cognitive approaches, and would want to see these two aspects integrated into an overall

framework. Similarly, there is no inherent contradiction between corpus studies and cognitivism: indeed, the reverse is the case, as has been persuasively argued by Schönefeld (1999). The feasibility of a contribution from corpus linguistics to cognitive studies is demonstrated by work such as that of Leech, Francis and Xu (1994), who used computer-assisted analysis of corpora to show, textually, the gradience of linguistic categories.

In summary, then, I have considerable reservations about the idea that we should throw away our existing theories and build up descriptions of languages piece by piece, in the hope that the pieces of the jigsaw will eventually come together to reveal something approaching a theory of language. Despite these reservations, I remain convinced that if functional theories are to live up to the claim that what they are modelling is communicative interaction, then it is essential that they take fully into account the kinds of evidence constantly thrown up by corpus analysis. In the light of these observations, it is of interest to examine the ways in which textual data have actually been used in functional linguistics. It is to this that we now turn.

5. The use of textual materials in functional theories

The position of functional theories on a cline from 'most formal' to 'radically functional' correlates with the extent to which textual materials have been used in description and theory-building. Let us first consider the two ends of the spectrum. RRG has so far made very little use of authentic text, except to the extent (which is difficult to assess) that examples from the wide range of languages studied are taken from collections of attested material. In SFG, on the other hand, the study of text has always been a priority. This situation can be explained largely in terms of the goals of the theories concerned: as we have seen, RRG is strongly committed to the formulation of a theory with a high degree of typological adequacy, as evidenced by cross-linguistic syntactic, semantic and pragmatic comparisons, while SFG developed largely, though not exclusively, in the context of the study of a single language, English, and has always emphasised the importance of texts. True to the influence of Halliday's teacher, J. R. Firth, who insisted on constant 'renewal of connection' between theory and data, Halliday's (1985, 1994a) *Introduction to Functional Grammar*, the book through which the theory is probably best known, was written specifically with the needs of text analysts in mind. The theoretical approach which underlies this account is described as follows:

> In general, [...] the approach leans towards the applied rather than the pure, the rhetorical rather than the logical, the actual rather than the ideal, the functional rather than the formal, the text rather than the sentence. The emphasis is on text analysis as a mode of action, a theory of language as a means of getting things done. (Halliday 1994a: xxvii)

SFG is also very much concerned with the relationships between texts and the contexts in which they are produced and understood, and is the only functional theory to have built in a specific model of social context, through the parameters of field, tenor and mode. It is not surprising, then, that systemicists have frequently used textual materials as a source of evidence for theory building. A similar situation obtains for WCF, which also has a strong textual orientation, as can be seen from, for example, the detailed work on topic continuity and referent tracking reported in Givón (1983), or the recent work of Thompson (2001, 2002) on conversational data.

FG is situated, with respect to textual studies, between the two extremes. The seminal work of Simon Dik shows very little evidence indeed of engagement with any kind of authentic material, and again we may explain this in terms of the strong typological orientation of the theory, and also the postulation of a largely unspecified 'theory of verbal interaction' which was considered quite separate from the grammar itself. On the other hand, there have always been two groups of scholars working within the FG framework, both centred on Amsterdam, the birthplace of FG, whose interest has been primarily in adducing evidence from texts. Firstly, a group of classicists, foremost among whom are Bolkestein, Kroon and Risselada, have worked on a range of topics including cohesion, information distribution (topic, focus, etc.), illocution and discourse particles, in Latin and Ancient Greek. Secondly, a group of anglicists, among whom Mackenzie, Hannay and Keizer are particularly prominent, have used texts in their work on information distribution in English. Mention should also be made of the work of de Vries on Papuan languages.[6] We shall see later that at the present time we are witnessing considerable growth in the use of authentic materials in FG.

In many cases, the use of textual materials in functional studies has been limited to isolated fragments, or at the most just a few short texts, chosen for their suitability for the purpose intended. Understandably, such studies are normally done manually rather than with computer assistance. A survey of the literature on SFG and WCF would reveal large numbers of examples of such work, including most of the work on textual structure (theme/rheme patterning, including thematic progression; information structure as signalled by

intonational means in English; cohesion; topic continuity and shift; and so on) which has been such an important feature of work in these two sets of approaches. A much smaller number of studies of this kind can also be found in the FG literature, largely in the work of the Amsterdam classicists and anglicists mentioned earlier.

In many other cases, scholars have put together their own collections of text for particular purposes, either collecting the data themselves or assembling components from various available sources, and although some attention is often given to questions of representativeness, these may have to take second place to what is available for the language or language variety under study. These collections thus vary widely in the extent to which they approximate to the definition of a corpus given in §3. The techniques used for searching the textual materials may be manual, computational or a mixture of the two. Into this category would fall a range of studies conducted by proponents of FG in the past few years. A considerable amount of work within SFG is also of this kind: much of Matthiessen's work, for example, makes use of what he refers to as a very large 'archive' of texts which, though gathered opportunistically from various sources, can be searched according to explicit criteria to isolate texts of particular types (see e.g. Matthiessen 2004).

Yet other work uses one or more already established corpora of texts.[7] One subgroup of such studies has used what today would be considered quite small corpora (1 million words or less), often indeed taking limited extracts from such corpora, and may employ either manual or computerised techniques, to select data in order to provide detailed accounts of a particular area. Once again, such work has been done within both FG and SFG. Other work uses the whole of a corpus (either one of the older ones, in the 1 million word range, or a very much larger and more recent corpus such as the COBUILD BofE or the BNC) as a source of data, taking systematic samples by automatic means if necessary because of the large number of examples potentially available; such work inevitably uses computerised searching techniques, though we shall see later that different studies use the corpus in rather different ways.

Finally, mention should also be made of the use of two rather special types of material which have been used in functional studies: collections of child language such as the CHILDES database, and collections from languages with a limited number of texts available, such as Biblical Hebrew.

Two things stand out in these types of study. Firstly, apart from the use of manually-analysed text fragments and/or small numbers of short though complete texts, the studies tend to be quite recent, and although large corpora

such as the BofE and the BNC have been available for some years now, less use has been made of such resources in functional linguistics than might have been expected. Secondly, even when computer-searchable corpora have been used, sometimes only a small sub-corpus has been manually analysed. It seems, then, that relatively few studies so far have used computerised searches of large bodies of text, and it is worth asking ourselves why this might be.

One probable reason for this low take-up is perceptions about the limitations of corpora themselves. Until very recently, there was a dearth of corpora of spoken language, and although this problem has now been partly addressed, for English at least, by the inclusion of spoken components within major corpora such as the BofE and the BNC, any analysis based on these materials soon reveals that they are heavily skewed in the direction of educated, often academic, usage.[8] This, of course, is just one aspect of the general problem of representativeness discussed earlier.

A further problem is that until recently, very few large corpora of languages other than English were available. For instance, a considerable proportion of the work on Spanish which will be reported on later in this paper was done using a corpus of about 150,000 words derived from a set of spoken materials, *El habla de la ciudad de Madrid* (Esgueva and Cantarero 1981). This has now been incorporated into the *Macrocorpus de la norma lingüística de las principales ciudades del mundo hispánico* (Macrocorpus of the linguistic norm of the main cities of the hispanic world) (Samper Padilla, Hernández Cabrera and Troya Déniz 1998), containing samples of similar size from 11 other cities in Spain and Latin America. Only more recently have larger corpora been assembled, such as the 130 million word Corpus de Referencia del Español Actual (CREA) created at the Real Academia, containing both written and spoken material, and the Corpus del Español elaborated by Mark Davies, with 100 million words including written texts spanning the 13th to the 20th centuries plus some spoken texts. The situation is thus improving for Spanish, as it is for many other languages: for instance, there are now national corpora of Czech and Hungarian, and similar corpora are in preparation for Croatian and Polish; furthermore, various corpus materials, often restricted in coverage, are available for a fairly wide range of European and Asian languages.[9]

Further limitations are concerned not with the corpora themselves, but with the computer-based methods which must be adopted if large amounts of language are to be analysed. Certainly, such methods are rapidly getting more sophisticated, but it remains true that the software which is readily available to researchers outside large commercially-financed projects is capable of performing

only a fairly restricted range of analyses, based on surface features of form: ultimately any computer-based analysis is dependent on recognition of sequences of characters in an electronically encoded representation of language, which means that the easiest operations are those involving word forms and punctuation. The fact that most corpora are tagged with word class categories does, of course, allow some grammatical analysis: for instance, it would an easy matter to search in such a corpus for a verbal form followed by two noun phrases, in order to study ditransitive complementation patterns. There is also a growing number of corpora which have skeletal syntactic parses attached, such as components of the International Corpus of English (ICE) and the SUSANNE Corpus (a 130,000 word sample derived from the Brown Corpus: see Sampson 1995), and this enhances still further the range of questions which can be addressed. It remains true, however, that many phenomena of interest to the functional linguist are still outside the scope of the computational techniques implemented in the available software. Particularly difficult to automate are searches for categories which are important in the study of information distribution and discourse structure, and this problem is, of course, particularly acute for those approaches, such as SFG and WCF, in which such issues have loomed large. The development of corpora with discourse annotation is recent and quite limited, and of course even these corpora may not be annotated for the categories of interest to particular theories. Unsurprisingly, then, even studies which have used materials such as the London-Lund Corpus of Spoken English (LLC) have often had to search manually, often in quite small amounts of text, for the features of interest to the study. A particularly promising development is Wu Canzhong's SysConc application, a tool which provides more information, within a systemic linguistic framework, than a conventional concordancer.[10]

Because computer-assisted analysis prioritises word forms, it is not surprising that the word has been the unit under focus in many studies of this kind. Indeed, the contribution of corpus linguistics to the production of a more adequate treatment of words in dictionaries has been spectacular (see e.g. Sinclair 1987, Landau 2001). This orientation does, however, lead naturally to a lexically-based approach, and this is no doubt seen by some as distorting the concept of a grammar. However, two further points are relevant here. Firstly, there has been a movement, in recent years, towards a more lexically-oriented view of language, in grammars occupying various positions on the formal/functional continuum (e.g. Lexical Functional Grammar, Hudson's Word Grammar, the branch of FG known as the Functional Lexematic Model).

Secondly, this shift does not entail the abandonment of syntax, but rather the recognition that syntax and lexis are strongly interdependent aspects of linguistic patterning.

An additional problem is the considerable investment of time and energy required to interpret the data produced by automatic analysis of even a small area of the lexicogrammar. Of course, the amount of data processed can be reduced, with consequent savings in time, but also with decreased confidence in the outcomes.

Finally, as we saw earlier in this paper, the results of studies on large corpora present very serious challenges which can strike at the very heart of traditional assumptions about language patterning and how we should describe it. This is clearly highly uncomfortable for many linguists, as it appears to require a radical rethinking of the ways in which we approach linguistic description and theorising. This sense of discomfort, compounded by the 'mechanical' nature of the initial analysis by computer, the 'messiness' of corpus data, and the sense of loss of control in the face of numbers of words running into the hundreds of millions, is probably one of the most important factors which is still holding some functional linguists back from using corpus analysis. For some functionalists, even full engagement with language in use seems threatening: despite Dik's (1997a: 6) rhetoric about studying language 'within the framework of the rules, principles and strategies which govern its natural communicative use', in practice he sanitises FG as a grammatical theory by separating it from an unspecified, unresearched 'theory of verbal interaction' (for discussion, see Butler 1991, 1999a, 2003).

The limitations of analysis based on corpora are real, and we need to be suitably cautious about our interpretations; on the other hand, if we are to take seriously our commitment to the study of natural language in communicative use, we must take on board those phenomena, revealed by the work of corpus linguists, in which we can have some confidence.

What, then, has actually been done with textual materials in functional theories? Limitations of space and my own expertise clearly preclude an exhaustive survey, so the discussion will be restricted in two ways: I shall look only at work carried out within the frameworks of FG and SFG,[11] and I shall concentrate on those studies which use textual materials that would qualify (to varying extents) as corpora under the definition given in §3. This means that I shall not give examples here of work, much of it very valuable, on collections or 'archives' of material assembled by individual scholars for the purposes of their work, neither shall I be concerned with the very large amount of work which

has used individual texts or small groups of texts. This inevitably entails that a large proportion of text-based work will not be covered here. Partly this decision is due to the dictates of space, but it is also a reflection of the basic orientation of this paper towards the challenges thrown down by corpus linguists on the basis of the study of 'large' corpora. The aim is to assess the extent to which work in functional linguistics has made use of corpus materials of various kinds, the kinds of purpose for which they have been used and, most importantly, the extent to which corpus data have been used to challenge central aspects of the theories themselves.

In some cases the role of the corpus is merely ancillary, in that it is used merely as a source of examples to illustrate and support the claims being made.[12] In the majority of cases, however, the primary aim is to provide a description of some area of the (lexico)grammar of a language in terms of a particular theory, using the corpus data to base that description on actual usage rather than on invented examples.[13] Studies such as these are important for functional theories, not only because they add to the description of particular phenomena within the theoretical framework, but also because certain details of the theory itself may need to be modified to accommodate the corpus data. It is, however, rather rare for such studies to lead to a radical reassessment of the fundamental principles and categories on which the theory is based: the aim is rather to fill in the detail within a framework whose basic architecture is taken for granted. Such studies are thus very typical of the kind referred to by Tognini-Bonelli and Sinclair as 'corpus-based'.

In yet other studies, attested data are used in order to test predictions made on the basis of a functional theory, as when Boland (1999) sets up a set of hypotheses concerning the order of acquisition of adverbs at different levels in the FG structure of the clause. To the extent that such hypotheses are not refuted, the work can be seen as providing provisional support for the theoretical statements from which the hypotheses were derived.

There are, however, recent studies which adopt a middle way between the kinds of study reviewed above and the corpus-driven approach, in that although they are usually based in some particular area of the (lexico)grammar, they test and indeed challenge the adequacy of fundamental theoretical concepts when faced with corpus data. Examples of such work include: Tucker's work on the relationship between lexis and grammar in SFG, in relation to semi-fixed multi-word expressions; Nuyts' study, within his Functional Procedural Grammar framework, of modality in Dutch and German; and my own work on the assessment of theoretical claims made in FG, SFG and RRG. I believe that

studies of this kind offer the best way of ensuring that corpus studies have a direct impact on basic theoretical constructs in functional linguistics, and I shall therefore look at each of them in rather more detail. Significantly, two of the three studies are largely concerned with how we can respond, within functional theories, to some of the challenges thrown down by corpus-driven linguists.

Tucker (1996) is concerned with the modelling of lexis in the Cardiff 'dialect' of SFG on which he has worked extensively, in collaboration with Robin Fawcett. More specifically, Tucker's paper shows how corpus evidence relating to the phraseology of semi-fixed expressions can be accommodated within the systemic model. In Hallidayan SFG, lexis and grammar are conceived not as separate levels of patterning, but rather as part of a single 'lexico-grammatical' level: both grammatical and lexical phenomena are involved in the realisation of underlying meanings, and the difference between them is seen as one of delicacy, in that while grammatical patterns tend to realise the broader distinctions of meaning, lexical patterns tend to realise the finer distinctions. In other words, lexis is seen as 'most delicate grammar'. As Tucker points out, this view is in potential conflict with the corpus linguist's view, namely that the lexis drives the grammar, in the sense that grammatical patterns arise out of the complex behaviour of many individual words. Tucker's aim, then, is to reconcile these two seemingly opposed positions. He points out that some choices which get realised lexically are indeed at the rightmost, most delicate end of system networks: for instance, a number of systemicists have shown how the senses of sets of individual English verbs can be generated by means of networks pushed to high levels of delicacy. However, the choice of a particular lexical item can in turn create conditions which restrict the range of further options available, for example in complementation: as Tucker observes, the semantic difference between LIKE and DISLIKE could be modelled as a very delicate choice in a transitivity network, but the two verbs have different possibilities with respect to the complements they can take (*I like/dislike going for a walk every morning*, but *I like/*dislike to go for a walk every morning*). Tucker points out that in a theory which does not make a clear separation between grammar and lexis, any consequences of lexically-realised choices can be modelled in terms of systemic dependency, whether those further choices are realised structurally or lexically. He goes on to illustrate his position by means of a detailed account of how a systemic grammar can generate semi-fixed expressions of the type *I haven't the faintest/foggiest/remotest/slightest idea/notion.* What Tucker achieves, then, is a reinterpretation of a very central facet of Hallidayan systemic theory in terms which can accommodate apparently recalcitrant data from corpus

studies. In Tucker (2004) the arguments are generalised to a range of phenomena whose handling in the Cardiff SFG grammar has been informed by corpus data: not only fixed and semi-fixed expressions, but also collocational patterning more generally, and the probabilities attaching to options in systemic networks.

Nuyts (2001) presents a very careful and detailed computer-assisted analysis of modal adjectives and adverbs, as well as other means of expression such as modal verbs and mental state predicates, in a structured 1-million word corpus of Dutch and a similar corpus of German. He uses this analysis to argue convincingly against FG proposals by Hengeveld (1987, 1988, 1989) which postulate a distinction between objective and subjective epistemological types of modality and claim that modal adjectives are always objective, while modal adverbs are subjective. Instead, Nuyts is able to show that a convincing account of the complexities of modality in his corpus samples can be given in terms of a set of four properties: evidentiality, performativity, information structure in the discourse, and discourse strategy (mitigation, argument management). But even more importantly from the point of view of FG theory, he shows that the layering system which, in one form or another, is central to most FG accounts (at its simplest, a representational and an interpersonal level, each consisting of its own set of distinct layers), is far too rigid to accommodate the rich patterning revealed in a corpus study. Nuyts himself favours shifting much of the complexity to a pre-linguistic, conceptual level in his Functional Procedural Grammar model.

In my own work I have demonstrated, through corpus studies, the need for important modifications to fundamental concepts of FG and its offshoot, the Functional Lexematic Model. In mainstream FG, the predicate frame is the central structure from which the underlying structure of the clause is built up, and consists of a predicate variable, the form of the predicate itself, its type (e.g. verbal, nominal), the number of arguments and the semantic functions associated with these arguments. In work designed to show the need for at least a considerable expansion in the information incorporated into predicate frames,[14] I effectively take a corpus-driven perspective on a set of individual predicates in English and Spanish in attempting to provide a description which takes account of all the patterning found, then confronting in as open and honest a manner as possible the problems the data cause for the FG conception of the predicate frame. The verbs ORDER and CONSIDER in English, and PASAR (lit. 'pass') in Spanish, are considered in Butler (1998), the verbs GIVE and TAKE in Butler (2001). Inevitably, such work leads to the proposal that collocational and colligational information, also details of differential use in particular

varieties of a language, should be built into the information provided about a predicate in the lexicon, and a format for an enriched predicate frame, with links to schematic networks of senses, is suggested (Butler 1998: 189).

In other work (Butler 1997) I use Spanish corpus data to investigate a further aspect of collocational and colligational patterning which had not been addressed in FG, that of the multi-word sequences which, as we have seen, are so important, both qualitatively and quantitatively, in our use of language. These sequences are classified both structurally and functionally, and it is found that most involve the nominal area of the grammar (including the complements of prepositions), so giving some hope that a substantial improvement to FG could be gained by allowing for a phraseological component in this part of the grammar.

The aim of Butler (1999b), which takes verbs of light emission in Spanish as its area of exemplification, is to demonstrate the need for expansion of the Functional Lexematic Model to include not only syntactic syntagmatic relations of complementation, but also those of collocation and colligation. In Butler (2002) I present an analysis of the predicate CATCH, based on examination of three dictionaries, an informant test, and a detailed study of actual usage as reflected in the BNC. The paper considers the extent to which the findings can be accommodated within the Functional Lexematic Model and also the lexical grammar model advanced recently by Mairal Usón and his colleagues (Mairal Usón and Van Valin 2001, Mairal Usón and Faber 2002), which combines aspects of the FLM and of RRG. The conclusion reached is that although the adoption of RRG-based logical structure formulations permits us to give a quite insightful account of similarities and differences among the uses of forms of CATCH, no lexical template of the kind proposed in the model is adequate to capture all uses of the predicate, and that the properties common to all uses are shared by other verbs of capture.

Finally, there are many points in my detailed comparison of FG, RRG and SFG (Butler 2003) where corpus data are used to challenge claims made in these theories, rather than simply to illustrate and support descriptions.

Even work of the kind I have described above does not, of course, go by any means all the way in responding to the challenge of the 'lexical items' or 'extended units of meaning' which emerge so clearly from corpus-driven work. One important issue is the need to accommodate extended units of meaning of the type 'visibility + preposition + *the* + *naked* + *eye*, plus the overall semantic prosody of difficulty', within the constituent-based framework of almost all grammars. There are those who believe that there is no need to reconcile the two positions, in that the grammar can be seen as a statement of the static

potential of a language, while the idiom principle comes into play when that potential is actualised in communicative events. However, for those of us who believe that the whole grammar, in the widest sense, should be a model of the native speaker's production and understanding of language, this dual option is not available. If we are to give a processing-oriented account of the grammar, then it must, logically, all fit together into a compatible set of procedures, acting on a compatible set of categories.

In this light, it is interesting to observe that a number of models have recently emerged, under the general umbrella of FG theory, which are intended to increase the psychological/cognitive adequacy of the theory by means of a processing orientation. I have already mentioned Nuyts' Functional Procedural Grammar in connection with his work on modality: this model (see Nuyts 1992, 2001: 272–279, 2004) is intended to capture the cognitive structures and processes involved in language production. Hengeveld (2004a, 2004b) has proposed a Functional Discourse Grammar which, although intended to model product rather than process, can be implemented by means of a staggered activation of the levels it proposes, in order to simulate production. Mackenzie (2000, 2004) has proposed an Incremental Functional Grammar based on the idea that the basic form of utterance is the holophrase, which can be expanded into larger, though still not necessarily 'complete', forms. Bakker and Siewierska (2004) have discussed a model of the language producer in which production occurs in a top-down, depth-first manner and due consideration is given to matters such as memory storage. As shown in Butler (2004), all of these models can accommodate to some extent the kind of multi-word item envisaged in terms of Sinclair's idiom principle, though many important aspects remain to be worked out.

A further problem is concerned with the interaction between the study of multi-word units and the typological adequacy criterion of functional grammars such as FG and RRG. If, as Wray claims, multi-word formulaic items are an important part of how our language processing mechanisms work, then we may confidently expect to find them in all languages. However, it remains to be seen how such expressions fit into the grammars of languages of different types. Progress in this area will be dependent on the availability and exploitation of corpora in a wide range of typologically different languages.

I will conclude this brief survey by mentioning three very recent papers which are particularly important in that they explicitly address, within the context of SFG, the issue of the challenges posed for linguistic theorising by the conclusions to be drawn from corpus analysis.

Hoey (2004) suggests that although SFG offers a more sophisticated account of relationships between grammar and lexis than most other theoretical approaches, it has still not proved able to represent vocabulary choice in ways which reflect the corpus-derived facts of collocation and colligation. He presents two theoretical positions for debate, though he retreats from both. The first is that grammar and lexis are separate, but related in that lexis, rather than being dependent on grammar, determines which grammatical features are most probable, and conditions the most refined grammatical choices. In this way of looking at things, grammar is part of the description of the word. The second possible view is that grammar is simply an epiphenomenon of the complete set of collocations and colligations shown by lexical items.

Hunston (2004), basing her arguments on illustrative analyses of the BofE, proposes some points of contact between SFG and the pattern grammar outlined in §4 of the present paper: both are seen as data-driven, and both attempt to account for grammatical choice as a socially-motivated phenomenon. However, she also perceives important differences between the two approaches. While SFG prioritises the paradigmatic axis of patterning, pattern grammar brings the syntagmatic axis back into focus, though it also tries to relate it to paradigm. A second area of potential conflict is concerned with the relationship between lexis and grammar. Hunston takes up Halliday's (2001) suggestion that pattern grammar might provide a link between the less delicate grammatical end of the lexicogrammatical cline and the more delicate lexical end. She also examines evidence for the more radical approach suggested by Sinclair's (1991b: 100) comment that "grammatical generalisations do not rest on a rigid foundation, but are the accumulation of the patterns of hundreds of individual words and phrases", in other words the second position discussed by Hoey (2004). Hunston's overall conclusion is that SFG and pattern grammar are parallel rather than conflatable approaches, and that it is useful to emphasise the complementarities between the two.

Finally, Stubbs (2004) offers us a very stimulating discussion of three types of unanswered question regarding the relationship between corpus studies of phraseology and theories within different paradigms, and the wide relations between these issues and cognitive and social patterning. The easiest questions are descriptive, and are concerned with the problem, noted earlier in the present paper, of generalising from the phraseological patterning of individual words to the language as a whole: here, Stubbs cites the pattern grammars discussed earlier, and also data from his own study of the strength of attraction between collocates in a 200 million word corpus derived from the BofE (Stubbs 2002).

Second level questions are concerned with what is basically the subject matter of the present article, i.e. whether corpus-derived accounts of phraseology can be reconciled with models proposed in other theories. In this connection, Stubbs suggests that links can be made between semantic preference/prosody and componential analysis, that corpus studies can provide a basis for a cognitive semantics, and that Sinclair's extended units of meaning share with speech act theory their emphasis on functionality and on the importance of the speaker. The most difficult, third level questions are to do with whether we can relate all the foregoing to social and cognitive theories (see also Stubbs 2001: Chapter 10). As an example, Stubbs suggests that corpus data could easily be used to collect evidence bearing on Searle's (1995) proposals for the construction of social reality.

The papers by Hunston, Hoey and Stubbs thus raise a number of issues which are crucial to any attempt to make functional theories sensitive to the data provided by corpus studies. The fact that these three papers leave the key issues largely unresolved is a useful reminder that we are only at the very beginning of this important debate.

6. Conclusion

The corpus-driven approach to linguistic investigation, which advocates the construction of descriptions and the building up of theory on the basis of what the corpus tells us, with minimal use of preconceived theoretical ideas, has certainly revealed a great deal of important information about the way in which naturally occurring language is organised. In particular, it has demonstrated that much of the language we use conforms to the idiom principle, according to which we make predominant use, in normal processing, of semi-prefabricated chunks of language, the open choice grammar having a much more limited role than that which has traditionally been attributed to it. This conclusion is also supported by Wray's psycholinguistically-oriented work on formulaic language. If functional linguists are to be true to the claim that they are concerned with language as communication, then they must take on board these important findings.

However, *pace* the criticisms of the corpus-driven linguists, I do not think that functional theorists should abandon the theories they have constructed, out of other kinds of evidence, over the years. There is far too much of great value in these theories for that. We must not, though, simply brush the results of corpus-driven linguistics under the carpet, regarding them as some kind of

optional add-on to the real business of constructing functional theories. Rather, we need an ambitious research programme in which corpora are used to tackle head-on the difficult task of investigating which aspects of our theories, including the most fundamental, we genuinely need to abandon in the face of the new evidence, which of them can remain, but with perhaps quite extensive modifications, and which are still robust enough to withstand the onslaught with only minor changes. As Tucker so aptly puts it:

> Corpus linguistic investigation is not a panacea for all the problems of linguistic theorising and description. It is unquestionably a valid point of departure and a sound basis, upon which to develop it, and against which to test theoretical and descriptive statements. (Tucker 2001: 212)

This testing has already begun, but needs to be developed on a much larger scale. I very much hope that in the near future we shall see continued expansion in the availability of reference corpora for a much larger range of typologically diverse languages, and much more use of such corpora by functional linguists to test even the fundamental pillars on which our theories are constructed, in an attempt to bridge a divide which some, too pessimistically in my view, have deemed unbridgeable.

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Notes

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1. Today, the 'West Coast' label seems rather out of date, since many proponents of this rather loose collection of approaches no longer work in the United States, let alone on the West Coast. I shall nevertheless continue to use it for convenience.

2. As Popper (1959:42) reminds us, "it is always possible to find some way of evading falsification, for example by introducing ad hoc an auxiliary hypothesis, or by changing ad hoc a definition".

3. The authors do, however, challenge in some respects the adequacy of traditional word classes, and add some new ones of their own, principally subclasses of noun and verb.

4. This is Halliday (1994a) in the bibliography for the present article.

5. Recently, Halliday and Matthiessen (1999) have discussed cognition in relation to systemic theory, but their view turns the usual position on its head, by claiming that cognition should be seen in terms of language, rather than *vice versa*.

6. Further details of these various types of study can be found in Butler (2003).

7. Obviously, this category overlaps with the previous one to the extent that researchers putting together their own set of textual materials have taken texts from existing sources.

8. Indeed, there are corpora specifically relating to this type of language, such as MICASE (the Michigan Corpus of Academic Spoken English) and CSPAE (the Corpus of Spoken Professional American English).

9. For lists of corpora in various languages, see, for example, David Lee's 'bookmarks for corpus-based linguists' at http://devoted.to/corpora and also Michael Barlow's corpus resources web site at http://www.ruf.rice.edu/~barlow/corpus.html.

10. At the time of writing, SysConc for the Macintosh platform can be downloaded from http://minerva.ling.mq.edu.au.

11. A survey of textually-based work in WCF would be a very valuable complement to that presented here, but I must leave this for those who have a detailed knowledge of this body of work. It is interesting to note, however, that the recent work of scholars such as Thompson (2001, 2002), and Bybee (Bybee and Scheibman 1999, Bybee 2001) using corpora of spoken American English, has adopted a 'usage-based' position which is somewhat similar to that of the corpus-driven linguists, in that grammar is conceived as emergent, arising out of combinations of reusable fragments.

12. Examples include: Verstraete (2001) [ideas from FG and SFG, modality, BofE/International Corpus of English (GB)]; Mairal Usón and Faber (2002) [Functional Lexematic Model and RRG, lexical templates for verbs of cutting, some exemplification from BNC]; Halliday and Matthiessen (2004) [SFG, detailed exposition of the grammar of English, range of corpus and other textual materials]; Butler (2003) [FG/RRG/SFG, detailed comparison of theories, BNC/LLC/various corpora of Spanish, Dutch, Gooniyandi].

13. Typical examples of work on English include: Halliday and James (1993) [SFG, tense choices, BofE]; Downing (1996a) [SFG, *get*-passive, BofE]; Halliday (1998) [SFG, the grammar of pain, BofE]; Tucker (2001) [SFG, the modal adverb *possibly*, bank of English/BNC]; Tucker (2002) [SFG, verbs of sun-tanning and burning, BofE]; Neale (2002, 2004) [SFG, transitivity, materials derived from BofE]; Downing (2001) [ideas from SFG, the modal adverb *surely*, BNC]; Carretero (1995, 2002) [SFG, epistemic modality, LLC]; Romero Trillo (1994a) [SFG, continuative elements, LLC]; Romero Trillo (1994b) [SFG, the role of filled pauses in thematic structures, LLC]; Herriman (2000) [SFG, extraposition in relation to text type, LOB]; Herriman (2003) [SFG, interpersonal functions of *wh*-clefts, Freiburg-

LOB (FLOB)]; Pérez Quintero (2002) [FG, adverbial clauses, LOB]; Gómez-González (2001) [SFG, but also other functional and cognitive approaches] Lancaster IBM Spoken English Corpus]. Examples of work of this kind on Spanish include: Butler (1992) [FG, relative clauses in spoken language, *El habla de la ciudad de Madrid*]; Downing (1997) [FG, discourse pragmatic functions of the Theme constituent in spoken language, *El habla de la ciudad de Madrid* plus material collected by the author]. Contrastive work on English and Spanish includes: Downing (1996b) [SFG, 'past-in-present' (perfect), *El habla de la ciudad de Madrid*/LLC]; Romero Trillo (1998) [SFG, continuative elements in discourse, *El habla de la ciudad de Madrid*/LLC]; Martínez Caro (1998, 1999) [FG, focus and emphasis, *El habla de la ciudad de Madrid*/LLC]; Lavid (2002, 2004) [SFG, theme/information focus, BNC Sampler/CREA/Macrocorpus]; Lavid (2001), Lavid and Arús (2002), Lavid, Arús and Zamorano (2002) [SFG, range of grammatical areas, BNC Sampler/CREA/Macrocorpus]. See also van Schaaik (1996, 2001, 2002) for FG-based descriptions of various areas of Turkish grammar, on the basis of a 6 million word corpus.

14. For further work, not derived from corpus studies, which also challenges the validity of the predicate frame as generally understood in FG, see García Velasco and Hengeveld (2002).

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